

MIDHAT SAJAD

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INTRODUCTION

Highly skilled and motivated Tunnel Engineer with 4+ years of industry experience in designing & analyzing complex geotech and tunnel engineering problems. Currently pursuing PhD in Underground Construction and Tunnel Engineering (UCTE) at Colorado School of Mines.

EDUCATION

- Colorado School of Mines, Golden, CO - (PhD in Underground Construction and Tunnel Engineering)** Aug 2024 - Present
GPA: 4.00
- Colorado School of Mines, Golden, CO - (Graduate Certificate in Underground Construction and Tunnel Engineering)** Dec 2025
GPA: 4.00
- Indian Institute of Technology (ISM), India - (M.Tech in Tunnelling and Underground Space Technology)** May 2021
GPA: 3.93 (Gold Medalist)
- University of Kashmir, Srinagar, India - (B.E in Civil Engineering)** April 2018
GPA: 3.92

TECHNICAL SKILLS

- Flac3D, RS2, S/Unwedge, slope/w, Midas Civil, STAAD Pro, Dlubal RFEM5/RSTAB9, Strand-7, Niosh, AdsecOaSys, Rhino, AutoCAD

RELEVANT EXPERIENCE

ARUP, San Francisco, CA – Tunneling and Underground Space Intern June 2026 - Present

- Involved in BART CP2 Project in California and Bloore Yonge Project in Canada

WSP in the US, Lakewood, CO – Geotech and Tunneling Intern May 2025 – May 2026

- Assisted in developing GBR Reports, GBR schedules and timelines to support geotechnical analysis for major tunneling projects, including the MWRA Boston tunnel, I-94 and contributed to precast/segmental tunnel lining design for other tunnel projects.
- Supported micro-tunneling (trenchless) projects in Canada, US by performing jacking force calculations for vertical curve alignments. Worked on preparing memos regarding RFP, RFQ, GBR and GDR.

Colorado School of Mines, Golden, CO - Teaching Assistant Aug 2024 – May 2026

- Working with Prof. Priscilla Nelson and Prof. Jamal Rostami for my PhD thesis and in Earth Mechanics Institute (EMI)
- Teaching Assistant for MNGN321: Intro to Rock Mechanics course/lab and course MNGN404: Tunnelling.

Larsen and Toubro, EDRC- Metro and Railways Business, India – Assistant Engineering Manager Aug 2021 – Aug 2024

- Experienced Design Engineer and Manager with a successful track record at L&T's EDRC.
- Proficient in rock-support interaction analysis, ground improvement solutions, slope stability analysis, designing tunnel structures.
- Successfully managed major metro and railway projects, including pre-bidding works, technical deliverables, and on-site issue resolution. Adept at BOQ preparation and stakeholder interfacing. Worked on major national and international projects.

Al-Farouk for Engineering Works Co., Egypt - Intern Dec 2019 – Jan 2020

- Planned, designed, and scheduled the Underground Optical Fiber Network for Nasr City, Cairo
- Managed the laying of optical fibers using various excavation techniques tailored to site conditions & financial constraints.

JK Engineering Reconstruction Agency, India - Trainee Feb 2016 – Mar 2016

- Gained hands-on experience with various construction techniques, including pier construction of flyovers.
 - Developed a comprehensive understanding of industry-standard concrete testing procedures.

RESEARCH EXPERIENCE

Boreability prediction model of TBM based on specific energy and machine parameters, India May 2020 – May 2021

- Improve TBM design for hard and abrasive rock masses, focusing on the Head Race Tunnel (HRT), Parbati Stage II, NHPC Ltd.
- Address the risks & cost implications of TBM excavation under unfavorable conditions compared to conventional methods.
- Develop TBM selection methodology and define major design and operational parameters to enhance performance in challenging rock mass settings.

Geometric and structural up-gradation of existing heritage boulevard road, India Apr 2017 – Mar 2018

- Upgrading the boulevard road in Srinagar, a key tourist attraction with high traffic volume causing congestion and discomfort. Situated on the outskirts of Dal Lake and surrounded by hills.
- Improve the road geometrically and structurally to alleviate traffic issues and address existing drainage problems.

PROFESSIONAL ASSOCIATIONS

- Women in Tunneling (WIT)
- Underground Construction Association (UCA)
- Society for Mining, Metallurgy and Exploration (SME)
- Society of Women Engineers (SWE)
- Rock Mountain Chapter: North American Society of Trenchless Tunnelling (RMNASTT)
- Tunneling Association of Canada (TAC)